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CHM 315 — Past Questions & Materials
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Course: CHM 315
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 300

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Paper 1 of 1 — CHM 315 POLYMERIC MATERIALS AND UTILIZATION OF WASTE 2024I2025

Level 300 • First Semester • PastQuestion

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DEPARTMENT OF CHEMISTRY
COLLEGE OF SCIENCE
FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
FIRST SEMESTER EXAMINATION 2024/2025 ACADEMIC SESSION
COURSE TITLE: POLYMERIC MATERIALS AND UTILIZATION OF WASTE
COURSE CODE: CHM 315

INSTRUCTION: ANSWER ANY FOUR QUESTIONS TIME 2 HOURS

1. (a) Discuss the steps involved in paint production (10 mrks)
(b) List five (5) chemicals/ substances each that can be used as (i) paint binder (ii) paint solvent (iii) pigment (7.5 mrks)
2. (a) Explain the following terms with two (2) examples each: (i) Thermoplastics (ii) Thermosets (iii) Aminoplast (iv) Phenoplast (10 mrks)
(b) mention five(5) classes of plasticizers and their applications (7.5 mrks).
3. (a) What do you understand by fillers, mention two major classes and give examples each (5 mrks)
(b) Explain five(5) defects experienced when paints are applied to surfaces.(7.5 mrks)
(c) With appropriate examples, explain the functions of extenders in a typical paint formulation (5 mrks).
4. (a) Briefly explain the following: i. Rubber tapping ii. Mastication iii. Calendering iv. Vulcanization. (8 mrks)
(b) In the synthesis of neoprene, the following reaction was carried out. Give the missing reagents and product.



2-chloro-1,3-butadiene

(3.5 mrks)

- (c) With chemical equation only, show the synthesis of Nylon-6,6 (6 mrks)
5. (a) Briefly explain the following: i. Saponification ii. Detergent iii. Cationic detergents iv. Anionic detergents. (6 mrks)
(b) When soap is added to hard water, magnesium ions of hard water react with soap forming insoluble magnesium salt of fatty acids. Use the reactants below to give the product of this reaction.



Soap

Hardness
of water

(4 mrks)

- (c) State five (5) differences between Soap and Detergents. (7.5 mrks)

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